PYTHON PROJECT · VERSION 2

BLOODCONNECT+

Finding the right blood, the right donor and the right help, in minutes, not hours.

Extended with blood availability, emergency nurse dispatch and a rare-blood registry

PRESENTED BY KANISHIK A



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THREE DELAYS MAKE AN EMERGENCY HARDER.

Finding blood, finding help and finding rare blood all take too long.

1

Blood is hard to locate

Donor details sit in notebooks and chats. Nobody can see which blood bank or hospital has stock of a given group right now.

2

Help is far away

Families lose time arranging transport and medical help. A trained nurse nearby could start care at home.

3

Rare blood is missing

Groups such as Bombay (hh) are often reported as common types in routine tests, and matching donors are very few.

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ONE APP CONNECTS THE WHOLE RESPONSE.

BloodConnect+ connects patients, donors, blood banks and nurses in one app.

01

Register

Donors, nurses, blood banks and patients create profiles with blood group and location.

02

Request

The patient raises a request with blood group, units needed and urgency level.

03

Match & respond

Nearest available donor, blood bank stock and nurse are matched and alerted.

04

Predict

Past requests are used to forecast which group will run short next.

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VERSION 1 PROVES THE MATCHING CORE.

A working Python prototype is the foundation for everything that follows.

Donor registration

Guided form to add name, phone, blood group and city.

Emergency search

Filter available donors by group and city.

Availability update

Mark donors available or unavailable.

Persistent storage

Records saved in a local JSON file.

Built with Python + JSON + command line

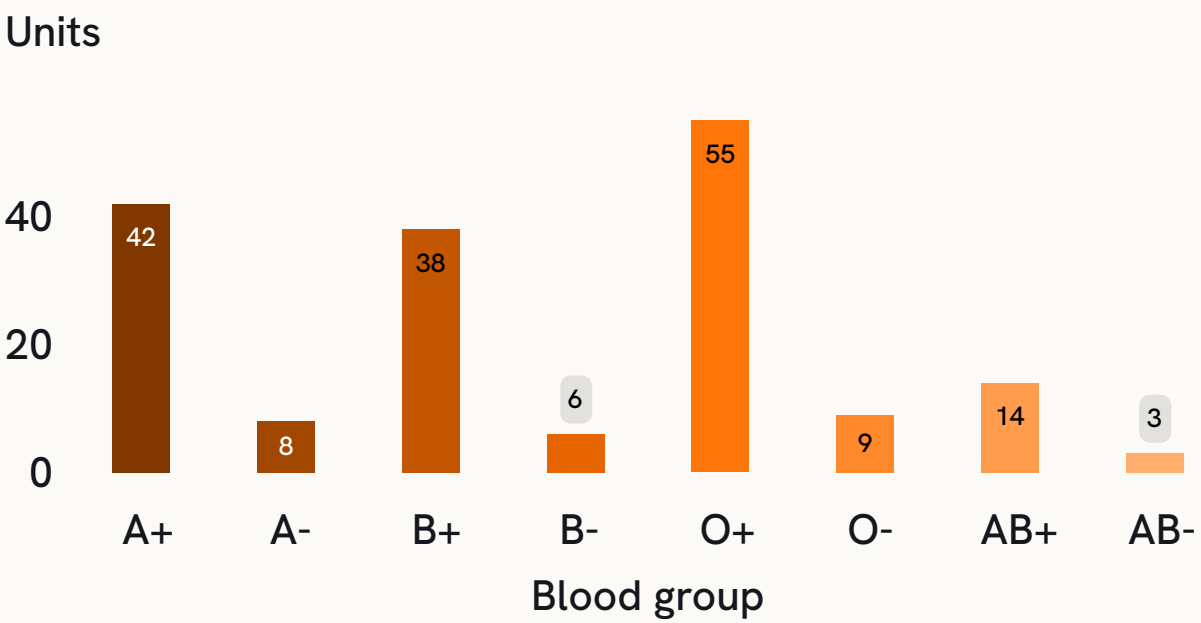
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05 · NEW CAPABILITY

SEE BLOOD STOCK BEFORE YOU TRAVEL.

Anyone can see which blood groups are available, and where.

SAMPLE DEMO DATA · ONE BLOOD BANK



Sample stock values for the demo, not real stock.

WHAT USERS SEE

- Stock by blood group at nearby banks.
- Low-stock warnings in red.
- Distance and a call button for each bank.
- A donor list as backup when stock is zero.

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ONE REQUEST STARTS TWO RESPONSES.

Blood matching and help move forward at the same time.

1

Patient raises request

Blood group, units, location and urgency.

2

System checks stock

Nearest blood bank availability is checked first.

3

Alerts nearest donors

Eligible donors nearby are matched and alerted.

4

Nurse is assigned

The nearest available nurse receives the request.

Nearest matches → tracked until the patient is helped

Result · help is on its way while blood is still being arranged

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A NURSE CAN ARRIVE WHILE BLOOD IS BEING FOUND.

The nearest available nurse is assigned, like a ride-booking app.

How it works

Verified nurses mark themselves available. The app picks the closest one and sends a request with the patient address.

Where it helps

Home care after accidents, post-donation weakness, elderly patients and emergencies where an ambulance is delayed.

What the patient gets

Nurse name, arrival time estimate and a call button. Status updates until the visit is complete.

Safety note

Prototype for coordination only. Real use needs verified nurses, hospital partnership and medical licensing.

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08 · RARE BLOOD

RARE DONORS NEED A REGISTRY, NOT A CITY FILTER.

Our most unique idea: a national registry for rare blood groups.

Bombay group (hh)

Very rare in India. Basic tests often show it as group O, so patients can receive the wrong match.

Rh-null

Among the rarest blood types in the world. Donors worldwide are very few, so every one must be registered.

How the registry works

Rare donors opt in. Searches go nationwide, not by city. Matching donors get an instant alert.



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START SIMPLE WITH PYTHON, THEN GROW.

Low cost, easy to demo, and the same code scales to a real app.

Python

Core logic, matching, distance calculation.

SQLite

Database for donors, stock, nurses and requests.

Flask / Streamlit

Simple web app that works on a phone.

SMS alerts

Simulated in demo, real service later.

JSON → SQLite · command line → phone · simulated → real service

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BUILD THE USEFUL CORE FIRST. GROW INTO A SMART NETWORK.

Roadmap to 2030. Future scope, not promises.

1

Now

Prototype

SQLite, blood stock search, requests and nearest donor matching.

2

Next

Emergency care

Nurse dispatch and rare-blood registry with alerts.

3

2028

Real-world pilot

Mobile app, real SMS, partner hospitals and blood banks.

4

2030

Smart network

Shortage prediction with AI, hospital system links and faster blood delivery.

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